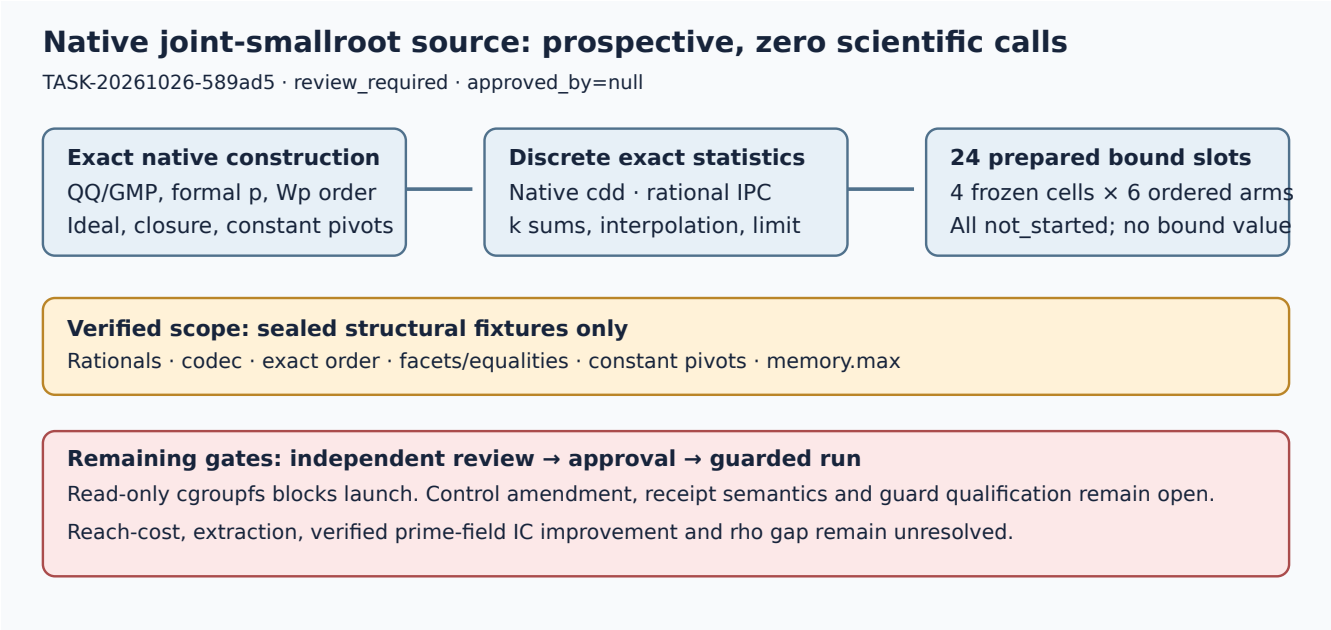
Native weighted joint-smallroot source

TASK-20261026-589ad5 · BATCH-ee839d · 26 October2026

review_required · approved_by=null · zero scientific calls

Standalone C++17 source and the exact24-call plan are prepared. A warnings-as-errors build, eight sealed structural fixture groups, an exact memory.max codec and two missing/unapproved admission-codec rejection checks pass. No scientific bound, control, ideal/closure, interpolation, calibration, reduction or extraction ran. This is partial interface readiness, not cuso equivalence or an IC improvement.



Native source. The owner directory is crypto:experiments/joint-smallroot-native/, branch research/joint-smallroot-native-20261026, based on e8fcadf62cd0b674c40ebc4e76dfe88e26e71f5c. Only that directory and an additive today log are changed. Actual workspace-local Conductor T-1 reserves both scopes at epoch1; parent maintains its lease. Owner snapshot 1d4798826 is published in crypto PR2642; parent verified77 source Git blobs. No merge is claimed.

Exact QQ/GMP sparse polynomials, formal-p weighted order, signed Sage comparison, first stripped coefficient, Singular ideal/basis/normal-form source, rational facets, convex integer domain, FIFO closure, constant-pivot intersection, discrete augmented statistics and rational interpolation/limit are implemented. Sage10.9 explicitly maps wdeglex to Singular Wp; actual bounded exponent/codec fixtures pass. Full scientific semantics remain unverified.

Dependencies and scope. g++14.2.0; C++17; O2; all warnings treated as errors. Singular4.4.1, GMP6.3.0, cddlib1!0.94n and OpenSSL3.6.5 are pinned by actual package/library/binary hashes. Installed cddlib declares GPL-2.0-only; a separate independently written generic GPL2 native helper exchanges exact rational text with the LGPL3 cuso-derived core. The core does not link cddlib. This architecture is recorded for review, not a legal certification. No binary is committed and no Python research runtime is used.

Original goal remains open. Establish the joint reach-cost tradeoff, then verify improvement over a named frozen current prime-field IC baseline. DLP recovery, full charged cost, matched rho gap, subgroup/target identities and independent replay remain unresolved. Single-block full-tree V and V/3 values are reference conversions, never joint feasibility certificates.

Evidence boundary and next gates

Requirement	Actual status
Source/build	Implemented and compiled; scientific routines unexecuted.
Sealed software fixtures	Eight original groups plus exact memory.max codec pass; missing/ unapproved admission envelopes reject.
Full compatibility	Unverified. None-valued/zero-leading failure timing remains partial; ideal/ NF cross-version equivalence not tested.
24-call plan	All24 slots not_started; scientific metrics null. Exact canonical algebra serialization is prepared source, not invoked.
Supervisor/admission	Real future entrypoints prepared. External authoritative receipt interpretation and actual guard qualification remain partial.
Current host	infrastructure-not-started: cgroupfs read-only; writable delegated8GiB tree guard unavailable.
Cost retention	Exact stage artifacts and whole-call cgroup CPU/memory prepared; per-phase child CPU unavailable and explicit.
Publication/approval	Source published; independent review, preapproval, Director lint/card, exact runbook approval and archive remain pending.

Control workload amendment. Prepared D-sign code verifies the fixed F101 sign/tree/corruption fixture and proposes four fixed(r,mu) control-only ideal/domain/closure constructions with denominator-cleared101^mu shift divisibility. No bound/limit is computed by these controls. They are separate from24 cold bound calls and need explicit Coordinator adoption and a reviewed control budget. No control ran; failure stops all bound calls.

Protection. Native supervisor source requires at least12GiB host MemAvailable, exact8589934592-byte memory.max, stopped-child cgroup placement, whole-tree kill, memory.peak/events and a monotonic1800-second watchdog. Current host cannot install that guard. No RSS-polling or unguarded fallback exists. The envelope binds files but still trusts declared external approval/lint statuses; actual upstream receipt semantics need integration before any admitted launch.

Validation. Owner spec/checks retains exact successful/failed build commands and fixture records. Initial wrong-directory and header-ordering failures are preserved and corrected. The final source passes warnings-as-errors compilation and the sealed fixtures. Parent library run:4844 passed,18 missing-fixture failures,121 ignored. All18 same-binary rechecks pass after unchanged tracked-input restoration; no clean full-suite pass is asserted. The original expectation hash is 6325487a3caa90db9aeae4d7ecd630aee021e55e4c39686af7ebe7992c0cb046. Exact guard codec expectation: b7ce2ae8405e8145c41bc7e1f07e2214936461122c683084bdd7532c3f3c8995.

Literature. Pinned cuso fb2747d2105a8e48499bf10a3a717cf80c14d079, Sage10.9 comparator/order/ matrix/ideal source and the frozen design/lead were read statically. Ryan eprint2024/1577 PDF returned403; identified university dissertation chapter3 fulltext returned429 in this session. Full Algorithm3/Lemma7/ theorem applicability remains unread/unverified. No paper-version equivalence, novelty, new mathematical theorem or Lean proof is claimed.

Provenance. Managed session /root/native_smallroot_implementation_20261026: parent-selected gpt-6.1-sol/high, executor-implementation, published task-local overlay, Executor authority; no API catalogue probe, fallback or degradation.

Source-linked record. report.md; compatibility-map.json; trialplan.json; transfer-assessment.md; owner-source-manifest.json; prospective-packet.json; validation.json; owner spec/director-request.md. Historical scientific graphs remain unchanged because this task produced no scientific observation.